

Social Science Inquiry II

Week 9: Pre-analysis plans and preregistration

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Winter 2026

Loading packages for this class

```
set.seed(60637)
# For plotting:
library(ggplot2)
# library(devtools)
# devtools::install_github("wilkelab/ungeviz")
library(ungeviz)
library(ggribes)
```

Today's plan

- Why p-hacking and multiple testing are a credibility problem
- What a pre-analysis plan (PAP) is, and what goes in it [Chen and Grady \(2021\)](#)
- Worked example: our climate-framing pre-analysis plan [Offer-Westort et al. \(2026\)](#)
- How preregistration changes interpretation (confirmatory vs. exploratory) [Monogan \(2015\)](#)

P-hacking

P-values

Suppose $\hat{\theta}$ is the general form for an estimate produced by our estimator, and $\hat{\theta}^*$ is the value we have actually observed.

P-values

- A two-tailed p-value under the null hypothesis is

$$p = P_0[|\hat{\theta}| \geq |\hat{\theta}^*|]$$

i.e., the probability *under the null distribution* that we would see an estimate of $\hat{\theta}$ as or more extreme as what we saw from the data.

- Suppose we have some data, $(Y, X_1, X_2, \dots, X_K)$.
- Suppose the null distribution represents the truth.
- If we test one hypothesis, what is the probability that we will find something that is statistically significant at $p \leq 0.05$?
- If we test two unrelated hypotheses, what is the probability that we will find something that is statistically significant at $p \leq 0.05$?

- A: event we reject hypothesis 1 at $p \leq 0.05$
- B: event we reject hypothesis 2 at $p \leq 0.05$
- $A \perp\!\!\!\perp B$: the two events are independent
- $P[A] = 0.05$
- $P[B] = 0.05$
- $P[A \cup B]$? The probability we see event A OR B?

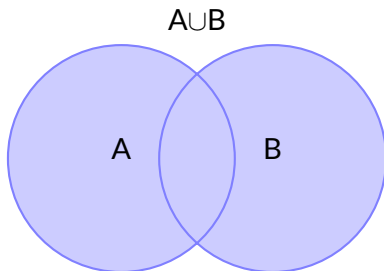
$$P[A \cup B] = P[A] + P[A^C] \times P[B|A^C]$$

A and B are independent, so $P[B|A^C] = P[B]$

$$P[A \cup B] = P[A] + P[A^C] \times P[B]$$

$$P[A \cup B] = 0.05 + 0.95 \times 0.05$$

$$P[A \cup B] = 0.0975$$



If the null is true ...

- and we conduct three independent tests, the probability that *at least one of them* will be statistically significant at $p \leq 0.05$ is
 $1 - 0.95^3 = 0.1426$
- and we conduct four independent tests, the probability that *at least one of them* will be statistically significant at $p \leq 0.05$ is
 $1 - 0.95^4 = 0.1855$
- and we conduct ten independent tests, the probability that *at least one of them* will be statistically significant at $p \leq 0.05$ is
 $1 - 0.95^{10} = 0.4013$

This becomes a real problem when researchers run many tests in their papers!

	Fail to reject null hypothesis ($p > 0.05$)	Reject null hypothesis ($p \leq 0.05$)
Null hypothesis true	True negative	Type I error, false positive
Null hypothesis false	Type II error, false negative	True positive

- **Type I error:** (false positive) we see an effect, where one doesn't really exist
- **Type II error:** (false negative) we didn't see an effect, but one really does exist

Multiple testing scenarios

- Comparisons across multiple treatments; A to B, B to C, A to C...
- Multiple outcomes
- Heterogeneous treatment effects (where is the cut point)
- Multiple regression specifications (specification search)

These tests aren't all fully independent, but the more tests we do, the more likely we are to uncover a false positive.

Ways to account for multiple testing

- Pre-specification of analyses
- Separating data in training and testing sets (more on this with machine learning)
- *p-value adjustment*

p-value adjustment

	Fail to reject null hypothesis ($p > 0.05$)	Reject null hypothesis ($p \leq 0.05$)
Null hypothesis true	True negative	Type I error, false positive
Null hypothesis false	Type II error, false negative	True positive

- Family-Wise Error Rate (FWER): the probability of falsely rejecting even one *true* null hypothesis; $P[\text{Type I error} > 0]$
- False Discovery Rate (FDR): expected proportion of false discoveries among all discoveries;
 $E[\# \text{ False discoveries} / \# \text{ All discoveries}]$

p-value adjustment

- Correcting FWER

- Bonferroni correction: for m hypotheses, for significance level α , implement α/m
- four independent tests, the probability that *at least one of them* will be statistically significant at $p \leq \alpha$ is $1 - (1 - \alpha)^4$
- For $\alpha = 0.05$, $1 - 0.95^4 = 0.1855$
- With Bonferroni correction: $1 - (1 - \alpha/4)^4 = 0.0491$
- Ten independent tests: $1 - (1 - \alpha/10)^{10} = 0.0489$

p-value adjustment

What if tests are not independent? Bonferroni is too aggressive.

$$P[AB] = P[A] + P[A^C] \times P[B|A^C]$$

If A and B are positively correlated $P[B|A^C] \leq P[B]$

- Correcting FWER
 - Holm correction: for m hypotheses, for significance level α :
 - Order the m conventionally calculated p-values from smallest to largest
 - Find the *smallest* p-value indexed as k such that $p_k > \frac{\alpha}{m+1-k}$
 - Reject all p-values greater than or equal to p_k , accept all p-values less p_k

p-value adjustment

- Correcting FDR
 - Benjamini-Hochberg correction: for m hypotheses, for significance level α :
 - Order the m conventionally calculated p-values from smallest to largest
 - Find the *largest* p-value indexed as k such that $p_k \leq \frac{k}{m} \alpha$
 - Reject all p-values greater than p_k , accept all p-values less than or equal to p_k

p-value adjustment

- In either case, for more complex settings, try simulation.

Multiple testing

- When can you consider tests as unrelated?
- Exploratory vs. confirmatory hypotheses?

What is a pre-analysis plan (PAP)?

Chen and Grady (2021)

- A PAP is a dated (often registered) document that specifies your design and analysis plan *before* analyzing outcomes.
- The goal is not “no exploration” — it is to make clear what was **confirmatory** vs. **exploratory**.
- A good PAP reduces researcher degrees of freedom:
 - outcomes and estimands are pre-specified
 - model/specification choices are pre-specified
 - inference rules are pre-specified (e.g., two-sided $\alpha = 0.05$, multiple-testing plan)

Why write a PAP?

Chen and Grady (2021) Monogan (2015)

- **Credibility:** reduces specification search and makes p-values/CI easier to interpret.
- **Transparency:** distinguishes confirmatory vs. exploratory work (both can be valuable).
- **Design discipline:** forces you to commit to outcomes, estimands, and decision rules before seeing results.
- **Scientific record:** helps counter publication bias by making “attempted” studies easier to find.

Confirmatory vs. exploratory

Monogan (2015)

- **Confirmatory:** the primary contrast(s), outcomes, and analysis choices were committed to in advance.
- **Exploratory:** additional outcomes, subgroup cuts, specifications, or hypotheses investigated after seeing data.
- Exploratory analyses are valuable — but they should be labeled as such and interpreted cautiously.

What goes in a PAP? (EGAP checklist)

Chen and Grady (2021)

1. Study design overview (unit, population, assignment, arms, outcomes)
2. Hypotheses (confirmatory vs exploratory; one- vs two-sided if relevant)
3. Measurement / index construction / transformations
4. Estimand(s) and estimator(s) (e.g., diff-in-means; regression adjustment)
5. Inference criteria (SE choice, α , multiple testing plan)
6. Procedures for data issues (missingness, outliers, attention checks)
7. Power / sample-size justification (assumptions explicit)

Where do you register a PAP?

Chen and Grady (2021)

- The key feature is a **verifiable time stamp** before analysis.
- Common registries (examples): EGAP Registry, OSF, AEA RCT Registry.
- Many projects also make a “computational PAP” by writing analysis code in advance and versioning it (e.g., GitHub).

Worked example: our climate-framing PAP (design)

Offer-Westort et al. (2026)

- Setting: survey experiment on climate policy attitudes
- Sample: ≈ 2300 U.S. respondents (Cloud Connect)
- Arms: 5 message frames + pure control; uniform random assignment
- Pre-test and post-test measured on the same policy-support items
- Outcome: an index (average) of 4 policy-support questions (scaled 0–3)

Worked example: our climate-framing PAP (analysis targets)

Offer-Westort et al. (2026)

1. Mean outcome by arm (overall and by party ID)
2. ATE for each frame vs. control (diff-in-means; plus regression adjustment)
3. “Best on average” frame vs. control (avoid selective inference via sample splitting)
4. “Best personalized” assignment vs. control (policy learning; sample splitting)

Best-arm selection: why sample splitting?

Offer-Westort et al. (2026)

- If you estimate 5 treatment effects and pick the largest, the usual p-value/CI for the “winner” is too optimistic.
- Sample splitting makes the selection step explicit:
 - **Training fold(s):** choose the best arm (by a pre-specified rule)
 - **Test fold:** estimate the effect of “being assigned the chosen arm” vs. control
 - Rotate folds so every observation serves as test data once (cross-fitting)

Deviations: what if your plan changes?

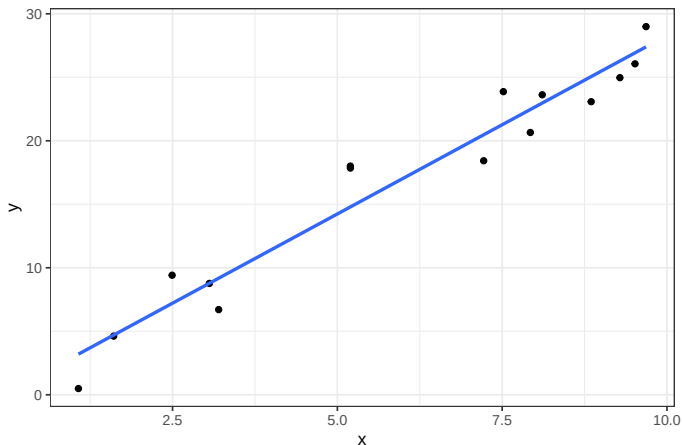
Chen and Grady (2021)

- Real studies rarely go perfectly (missingness, measurement problems, coding bugs, unexpected distributions).
- Good practice:
 - keep the preregistered analysis as the primary report
 - document and justify deviations (what changed, why, when)
 - clearly label additional analyses as exploratory/robustness checks

Some alternatives to confidence intervals (via ungeviz)

Show the underlying data.

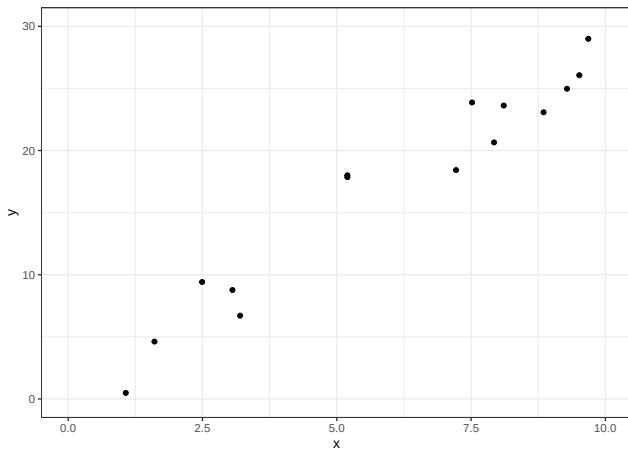
```
ggplot(iris, aes(Species, Sepal.Length, fill = Species)) +  
  geom_violin(alpha = 0.25, color = NA) +  
  geom_point(position = position_jitter(width = 0.3, height = 0), size = 0.5) +  
  geom_hline(aes(colour = Species), stat = "summary", width = 0.6, linewidth = 1.5, fun = 'mean')
```



Some alternatives to confidence intervals (via ungeviz)

Shaded confidence strips.

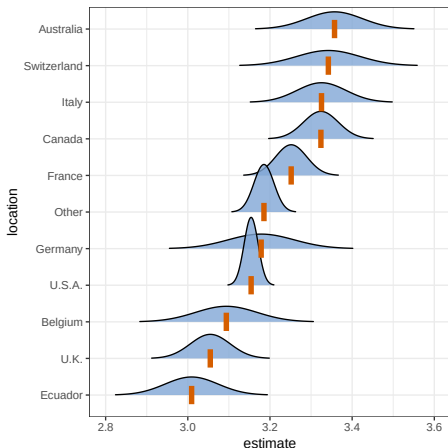
```
ggplot(cacao_means, aes(x = estimate, y = location)) +  
  stat_confidence_density(aes(moe = std.error), confidence = 0.68, fill = "#81A7D6", height = 0.7) +  
  geom_errorbarh(aes(xmin = estimate - std.error, xmax = estimate + std.error), height = 0.3) +  
  geom_vline(aes(x = estimate), linewidth = 1.5, height = 0.7, color = "#D55E00")
```



Some alternatives to confidence intervals (via ungeviz)

Confidence densities.

```
ggplot(cacao_means, aes(x = estimate, y = location)) +  
  stat_confidence_density(  
    aes(moe = std.error, height = after_stat(density)), geom = "ridgeline",  
    confidence = 0.68, fill = "#81A7D6", alpha = 0.8, scale = 0.08, min_height = 0.1) +  
  geom_vline(aes(x = estimate), linewidth = 1.5, height = 0.5, color = "#D55E00")
```



References I

Clause Wilke: <https://wilkelab.org/ungeviz/index.html>

Chen, L. and Grady, C. (2021). 10 things to know about pre-analysis plans. EGAP.

Monogan, J. (2015). Research preregistration in political science: The case, counterarguments, and a response to critiques. *PS: Political Science & Politics*, 48(2):425–429.

Offer-Westort, M., Gruen, W., Herron, C., Hyatt, K., Buford, M., Davis, K., Fonseca, D., Gurevich, M., Huang, T., Jerez, R., et al. (2026). Which frame fits? policy learning with framing for climate change policy attitudes. *Research & Politics*, 13(1):20531680251414927.